x_i	$d_i = x_i - \overline{x}$ $= x - 15$	d_i^2
11.4	−3.6	12.96
12.5	−2.5	6.25
12.8	−2.2	4.84
16.3	1.3	1.69
17.8	2.8	7.84
19.2	4.2	17.64
		$\Sigma d_i^2 = 51.22$

Standard deviation $\sigma = \sqrt{\dfrac{\Sigma d_i^2}{n}}$

$= \sqrt{\dfrac{51.22}{6}} = \sqrt{8.53}$

Hence, $\sigma \simeq 2.9$

(iii) Assumed Mean method

When the mean value is not an integer (since calculations are very tedious in decimal form) then it is better to use the assumed mean method to find the standard deviation.

Let $x_1, x_2, x_3, \ldots, x_n$ be the given data values and let $\overline{x}$ be their mean.

Let d_i be the deviation of x_i from the assumed mean A, which is usually the middle value or near the middle value of the given data.

$$d_i = x_i - A \text{ gives, } x_i = d_i + A \qquad \ldots(1)$$

$$\Sigma d_i = \Sigma(x_i - A)$$

$$= \Sigma x_i - (A + A + A + \cdots \text{ to } n \text{ times})$$

$$\Sigma d_i = \Sigma x_i - A \times n$$

$$\frac{\Sigma d_i}{n} = \frac{\Sigma x_i}{n} - A$$

$$\overline{d} = \overline{x} - A \text{ (or) } \overline{x} = \overline{d} + A \qquad \ldots(2)$$

Standard deviation $\sigma = \sqrt{\dfrac{\Sigma(x_i - \overline{x})^2}{n}} = \sqrt{\dfrac{\Sigma(d_i + A - \overline{d} - A)^2}{n}}$ (using (1) and (2))

$$= \sqrt{\frac{\Sigma(d_i - \overline{d})^2}{n}} = \sqrt{\frac{\Sigma(d_i^2 - 2d_i \times \overline{d} + \overline{d}^2)}{n}}$$

$$= \sqrt{\frac{\Sigma d_i^2}{n} - 2\overline{d}\,\frac{\Sigma d_i}{n} + \frac{\overline{d}^2}{n}\,(1 + 1 + 1 + \cdots \text{ to } n \text{ times})}$$

$$= \sqrt{\frac{\Sigma d_i^2}{n} - 2\overline{d} \times \overline{d} + \frac{\overline{d}^2}{n} \times n} \qquad \text{(since } \overline{d} \text{ is a constant)}$$

$$= \sqrt{\frac{\Sigma d_i^2}{n} - \overline{d}^2}$$

Standard deviation $\sigma = \sqrt{\dfrac{\Sigma d_i^2}{n} - \left(\dfrac{\Sigma d_i}{n}\right)^2}$

Thinking Corner

For any collection of n values, can you find the value of

(i) $\Sigma(x_i - \overline{x})$ (ii) $(\Sigma x_i) - \overline{x}$

8 Stats and Probability.indd 307 15-12-2020 03:24:11 PM